CS6140 MACHINE LEARNING

10 Practice Quizzes

78 new questions in the professor's five styles · answers after each quiz

Nguyen Ha · Midterm Thursday, Nov 12, 2026, 3:00 PM, Room 912

How to use. Do a quiz first **without looking**; then check the answer key that follows it. For every question you missed, the weak-spot table tells you exactly which page of the **Midterm Reference PDF** to re-read. Order = most likely on the exam first (topics already asked in class come first). Styles: (1) What/why · (2) Figure reading · (3) Compare & judge · (4) Mechanism · (5) Light math. Interactive version: **johnha.info/cs6140-midterm/**

Quiz order

#	Quiz	Why it is early in the list	Questions
1	W6 — Supervised Learning: kNN, Naive Bayes, SVM	4 in-class quiz topics (kernel, naive, zero likelihood, hinge)	8
2	W4 — Regression (linear, polynomial, regularized, logistic)	3 in-class quiz topics (n+1 points, k = 1/3/6, L1 vs L2 figure)	8
3	W7 — Unsupervised Learning: K-Means, DBSCAN, PCA	3 in-class quiz topics (K-Means convergence, PCA, scree plot)	8
4	W5 — Model Performance Evaluation	predicted numeric item (confusion-matrix metrics)	8
5	W12 — Intro to NLP (incl. LDA)	in-class quiz topic LDA (Latent Dirichlet Allocation)	8
6	W11 — Deep Learning	new 30-slide deck; perceptron / gradient step predicted	8
7	W8 — Decision Trees and Random Forests	predicted numeric item (Gini / entropy / information gain)	7

8	W9 — Ensemble Learning	compare-and-judge favorite (bagging vs boosting)	7
9	W10 — Recommender Systems	self-contained topic (not in Géron) — easy points if prepared	8
10	W2 — Math Review: Probability, Statistics, Linear Algebra	foundation; light Bayes / probability numbers	8

Quiz 1. W6 — Supervised Learning: kNN, Naive Bayes, SVM

Answer all questions first. Answer key on the next pages.

Q1 · STYLE 1: WHAT / WHY

A classmate says: “Naive Bayes is called naive because it ignores the class prior.” Which statement correctly fixes this?

A. It is naive because it gives every class the same prior.

B. It is naive because it assumes the features (words) are independent of each other given the class.

C. It is naive because it only works with two classes.

D. It is naive because it uses word order to compute likelihoods.

Q2 · STYLE 4: MECHANISM (HOW)

A new message contains the word “refund”, which never appeared in any VIP training post. (a) What happens to the VIP likelihood without any fix, and why? (b) How does the standard fix work?

Q3 · STYLE 5: LIGHT MATH

kNN with Euclidean distance. Query point (2, 2). Labeled points: P1 (5, 6) class A, P2 (2, 5) class B, P3 (1, 1) class B, P4 (4, 2) class A. What is the prediction for $k = 3$?

A. Class A

B. Class B

C. Tie — cannot decide

D. Class A because P1 is farthest

Q4 · STYLE 3: COMPARE & JUDGE

After training an SVM, you delete one training point that lies far from the hyperplane on the correct side (outside the margin). What happens to the hyperplane?

A. It moves toward the deleted point.

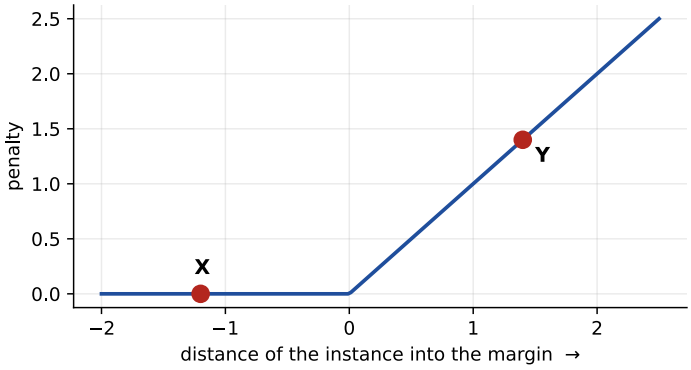
B. It does not change.

C. The margin becomes wider.

D. The kernel must be changed.

Q5 · STYLE 2: FIGURE READING

The plot shows the SVM penalty as a function of how far an instance is into the margin. Point X sits on the flat part, point Y on the rising part. For each: is it penalized, and is it a support vector?



Q6 · STYLE 5: LIGHT MATH

Hinge loss = $\max(0, 1 - y \cdot f(x))$ with $y \in \{-1, +1\}$. For an instance with $y = -1$ and $f(x) = 0.3$, what is the loss?

A. 0

B. 0.7

C. 1.3

D. -0.3

Q7 · STYLE 1: WHAT / WHY

Two features x and y cannot be separated by a straight line. Explain what the kernel trick does here, and how many dimensions a degree-2 polynomial kernel maps (x, y) into.

Q8 · STYLE 3: COMPARE & JUDGE

Features: annual income (dollars, 20,000–200,000) and age (18–80). Which model is MOST hurt if you skip feature scaling?

A. Decision tree

B. k-nearest neighbors

C. Naive Bayes on categorical words

D. Random forest

Answer key — Quiz 1

Q1 → B. The prior is used (the class's share of the training set, W6 p12). "Naive" = conditional independence: $P(\text{Message}|\text{Class}) \approx \prod P(\text{Word}|\text{Class})$, even though word order matters in real language.
Why the others are wrong — **A**: Priors are estimated from training proportions, not set equal. **C**: NB handles any number of classes. **D**: The opposite — it ignores word order.
Look up: W6 p11, p14 · Reference PDF p. 31

Q2 — model answer. (a) $P(\text{refund} \mid \text{VIP}) = 0/200 = 0$. The likelihood is a PRODUCT of word probabilities, so one zero makes the whole VIP likelihood 0 — VIP becomes impossible no matter how strong the other words are. (b) Laplace smoothing: add one fictional occurrence to every category's count, e.g. $(\text{count} + 1)/(\text{total} + V)$, so no probability is ever 0.
Look up: W6 p13 · Reference PDF p. 32

Q3 → B. Distances: $P3 = \sqrt{2} \approx 1.41$, $P4 = 2$, $P2 = 3$, $P1 = 5$. The 3 nearest are P3 (B), P4 (A), P2 (B) → majority vote = B.
Why the others are wrong — **A**: Only one of the 3 nearest (P4) is A. **C**: With $k = 3$ there is no tie (2 vs 1). Ties happen with even k — that's why odd k is preferred. **D**: Farthest points are not used at all.
Look up: W6 p6–p8 · Reference PDF p. 31

Q4 → B. Only support vectors (points on/inside the margin or misclassified) determine the hyperplane. A far, correctly classified point has no influence.
Why the others are wrong — **A**: Non-support vectors do not pull the boundary. **C**: The margin is set by the support vectors, which are unchanged. **D**: The kernel choice is unrelated.
Look up: W6 p16–p17 · Reference PDF p. 32

Q5 — model answer. X (flat part): penalty 0 — correctly classified and outside the margin; it does not influence the hyperplane (not a support vector). Y (rising part): positive penalty that grows linearly with its distance into the margin — it is inside the margin or misclassified, so it is a support vector and pushes the hyperplane.
Look up: W6 p19–p20 · Reference PDF p. 67

Q6 → C. $y \cdot f(x) = (-1)(0.3) = -0.3$, so $1 - (-0.3) = 1.3 > 0$ → loss 1.3. The instance is on the wrong side ($f(x) > 0$ but class is -1).
Why the others are wrong — **A**: Loss is 0 only when $y \cdot f(x) \geq 1$. **B**: That is $1 - 0.3$: you forgot the sign of y. **D**: Hinge loss is never negative (max with 0).
Look up: W6 p19 (formula from review file) · Reference PDF p. 34

Q7 — model answer. The kernel trick maps the data to a higher-dimensional space where a linear hyperplane CAN separate the classes; a linear boundary there is a non-linear boundary in the original space, computed efficiently by the kernel. Degree-2 polynomial on two features → 6 dimensions: x^2, xy, y^2, x, y, 1.
Look up: W6 p15, p18 · Reference PDF p. 32

Q8 → B. kNN adds raw differences inside the distance, so income (huge units) dominates age. Trees split one feature at a time on a threshold, so scale does not matter.
Why the others are wrong — **A**: Trees compare a single feature to a threshold — rescaling only moves the threshold. **C**: Category counts have no scale. **D**: A forest is many trees — same reason as trees.
Look up: W6 p6; W8 p13 · Reference PDF p. 34

Weak spots to review (tick the questions you missed)

Missed?	Q	Topic	Re-read
☐	Q1	Naive Bayes: meaning of "naive"	W6 p11, p14 · PDF p. 31
☐	Q2	Laplace smoothing / zero likelihood	W6 p13 · PDF p. 32
☐	Q3	kNN distance + majority vote	W6 p6–p8 · PDF p. 31
☐	Q4	Support vectors	W6 p16–p17 · PDF p. 32
☐	Q5	Reading the hinge-loss figure	W6 p19–p20 · PDF p. 67
☐	Q6	Hinge loss calculation	W6 p19 (formula from review file) · PDF p. 34
☐	Q7	Kernel trick	W6 p15, p18 · PDF p. 32
☐	Q8	kNN needs scaling	W6 p6; W8 p13 · PDF p. 34

*Summary line: if you missed ≥ 3 here, re-read this week's **Mechanisms** and **Comparisons** blocks before the next quiz.*

Quiz 2. W4 — Regression (linear, polynomial, regularized, logistic)

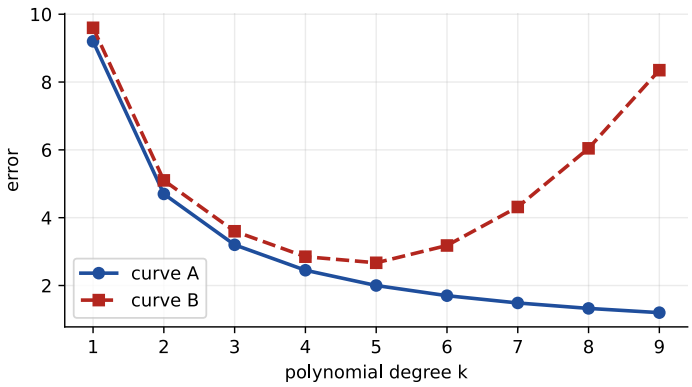
Answer all questions first. Answer key on the next pages.

Q1 · STYLE 1: WHAT / WHY

You have 8 training points. What is the lowest polynomial degree guaranteed to fit all 8 exactly, and why is that a red flag rather than good news?

Q2 · STYLE 2: FIGURE READING

The plot shows two error curves against polynomial degree. Which statement is correct?



A. A is test error; best degree is 9.

B. A is training error; B is test error; best degree is about 4.

C. B is training error because it is U-shaped.

D. Both curves say degree 1 is best.

Q3 · STYLE 3: COMPARE & JUDGE

You have 200 features but believe only about 10 really matter. Which regularizer is the better first choice?

A. Ridge, because it shrinks all weights evenly

B. LASSO, because the L1 penalty can set weights exactly to 0 (feature selection)

C. No regularization, because more features always help

D. Ridge, because it removes irrelevant features

Q4 · STYLE 5: LIGHT MATH

Gradient descent on $f(w) = (w - 3)^2$, starting at $w_0 = 0$, learning rate $\eta = 0.25$. What are w_1 and w_2?

A. $w_1 = 1.5$, $w_2 = 2.25$

B. $w_1 = -1.5$, $w_2 = -2.25$

C. $w_1 = 6$, $w_2 = 3$

D. $w_1 = 0.75$, $w_2 = 1.5$

Q5 · STYLE 4: MECHANISM (HOW)

In the deck's Ridge example, increasing α from 1 to 100 shrinks the slope from 570.63 to 95.90. What happens to bias and variance, and why must the features be standardized before using Ridge?

Q6 · STYLE 5: LIGHT MATH

In logistic regression, the log-odds $\ln(p / (1 - p))$ at some x equals $\ln 3$. What is p?

A. 0.33

B. 0.5

C. 0.75

D. 3

Q7 · STYLE 3: COMPARE & JUDGE

MAP estimation of logistic regression with a Gaussian prior $w \sim N(0, \sigma^2)$ is equivalent to:

A. MLE with no penalty

B. MLE plus an L2 penalty $\lambda w^T w$ (Ridge-like)

C. MLE plus an L1 penalty

D. A closed-form solution

Q8 · STYLE 1: WHAT / WHY

Give two of the deck's reasons why ordinary linear regression is a poor model for a 0/1 response.

Answer key — Quiz 2

Q1 — model answer. Degree 7 (it has 8 coefficients w0…w7 → 8 equations, 8 unknowns → exact solution). Zero training error happens even if the data is pure noise, so it tells you nothing about test error; the model memorizes noise = overfitting (deck: 'a degree-N polynomial exactly fits N+1 data points').
Look up: W4 p31 · Reference PDF p. 78

Q2 → B. Training error keeps falling as complexity grows (A). Test error is U-shaped (B): high on the left (underfit), rising on the right (overfit). Pick the minimum of B ≈ degree 4.
Why the others are wrong — **A**: A keeps decreasing — that is the training-error signature; degree 9 overfits. **C**: The U-shape is the TEST error signature. **D**: Degree 1 has high error on both curves (underfit).
Look up: W4 p31, p37; W5 p6 · Reference PDF p. 69

Q3 → B. LASSO's $|w|$ penalty drives some weights to exactly 0, performing feature selection and giving a sparse, interpretable model.
Why the others are wrong — **A**: Ridge shrinks but rarely zeroes weights — the 190 useless features stay in. **C**: Too many parameters relative to data → overfitting. **D**: That describes Lasso, not Ridge.
Look up: W4 p45, p48 · Reference PDF p. 21

Q4 → A. $f'(w) = 2(w-3)$. $f'(0) = -6 \to w_1 = 0 - 0.25(-6) = 1.5$. $f'(1.5) = -3 \to w_2 = 1.5 - 0.25(-3) = 2.25$. Moving toward the minimum at 3.
Why the others are wrong — **B**: That uses a PLUS sign (moving uphill). **C**: Forgot to multiply by η. **D**: Used $f' = w - 3$ (forgot the 2).
Look up: W4 p15 · Reference PDF p. 22

Q5 — model answer. Larger α → weights shrink toward 0 → simpler model → variance ↓, bias ↑ (the bias–variance tradeoff is controlled by α). The penalty is on weight size, and weight size depends on each feature's units, so without standardization features measured in small units get unfairly penalized.
Look up: W4 p41, p44, p48 · Reference PDF p. 21

Q6 → C. Odds $= p/(1-p) = 3 \to p = 3/(1+3) = 0.75$.
Why the others are wrong — **A**: That would be odds $= 1/2$. **B**: $p = 0.5$ when log-odds $= 0$. **D**: 3 is the odds, not a probability (probabilities ≤ 1).
Look up: W4 p59–p60 · Reference PDF p. 23

Q7 → B. The deck's MAP objective is $\Sigma \log(1 + e^{(-y\, w^T x)}) + \lambda w^T w$: the Gaussian prior adds an L2 penalty. MLE is MAP with an uninformative prior.
Why the others are wrong — **A**: That is MLE, i.e. MAP with an uninformative prior. **C**: An L1 penalty would come from a Laplace prior (not in deck). **D**: Neither MAP nor MLE has a closed form in general.
Look up: W4 p62–p65 · Reference PDF p. 22

Q8 — model answer. Any two of: (1) non-normal error terms; (2) non-constant error variance; (3) the response must satisfy $0 \leq p \leq 1$ but a line goes below 0 and above 1. → Use logistic regression (sigmoid keeps p in (0, 1)).
Look up: W4 p53–p55 · Reference PDF p. 21

Weak spots to review (tick the questions you missed)

Missed?	Q	Topic	Re-read
☐	Q1	Overfitting: N+1 points	W4 p31 · PDF p. 78
☐	Q2	Train vs test error curves	W4 p31, p37; W5 p6 · PDF p. 69
☐	Q3	Lasso vs Ridge choice	W4 p45, p48 · PDF p. 21
☐	Q4	Gradient descent step (minus sign)	W4 p15 · PDF p. 22
☐	Q5	Ridge: effect of α	W4 p41, p44, p48 · PDF p. 21
☐	Q6	Odds ↔ probability	W4 p59–p60 · PDF p. 23
☐	Q7	MAP vs MLE	W4 p62–p65 · PDF p. 22
☐	Q8	Why not linear regression for classification	W4 p53–p55 · PDF p. 21

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Quiz 3. W7 — Unsupervised Learning: K-Means, DBSCAN, PCA

Answer all questions first. Answer key on the next pages.

Q1 · STYLE 4: MECHANISM (HOW)

You run K-Means twice with different random initial centroids and get final objectives J = 412 and J = 389. Is the algorithm broken? Explain using how K-Means converges.

Q2 · STYLE 5: LIGHT MATH

1-D K-Means with k = 2. Points: 1, 2, 6, 8, 9. Initial centroids $c_1 = 1$, $c_2 = 6$. After ONE assignment + update step, the centroids are:

A. $c_1 = 1.5$, $c_2 \approx 7.67$

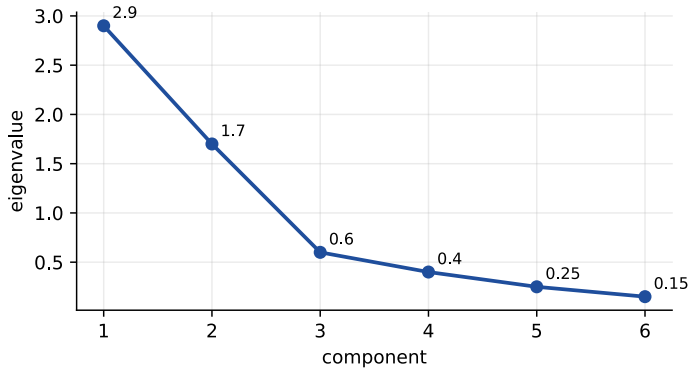
B. $c_1 = 1$, $c_2 = 6$

C. $c_1 = 3$, $c_2 = 8.5$

D. $c_1 = 1.5$, $c_2 = 8.5$

Q3 · STYLE 2: FIGURE READING

Scree plot for PCA on 6 standardized features (values shown). How many components should you keep, and about what share of the variance do they explain?



A. 1 component, 48%

B. 2 components, about 77%

C. 4 components, 93%

D. 6 components, 100% — always keep all

Q4 · STYLE 3: COMPARE & JUDGE

Data: two interlocking crescent-shaped groups plus a few isolated noise points far away. Which method fits best?

A. K-Means with k = 2

B. DBSCAN

C. PCA

D. K-Means with k = 3

Q5 · STYLE 1: WHAT / WHY

In PCA, what does a component's EIGENVECTOR tell you (as opposed to its eigenvalue)?

A. How much variance the component explains

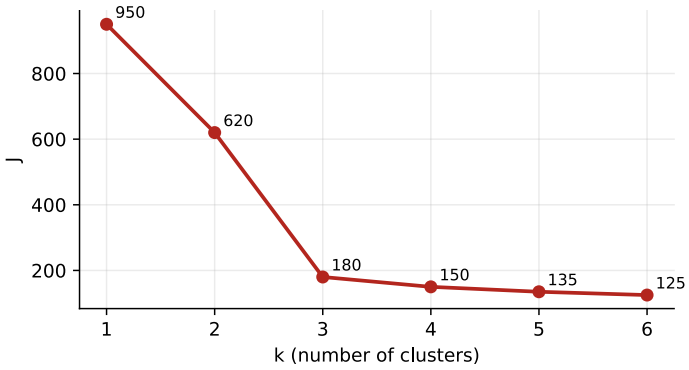
B. The coefficients of the original features that make up the component (its direction)

C. The number of clusters in the data

D. The p-value of the component

Q6 · STYLE 2: FIGURE READING

The plot shows the K-Means objective J for k = 1. . .6. Which k does the elbow method suggest?



A. k = 1

B. k = 2

C. k = 3

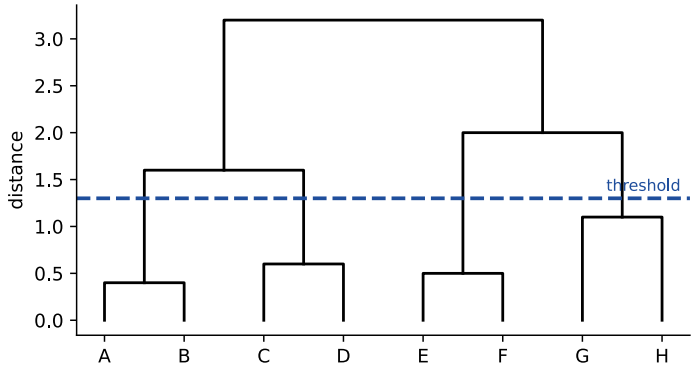
D. k = 6

Q7 · STYLE 4: MECHANISM (HOW)

DBSCAN with ε = 1 and min_samples = 4. Point A has 5 points in its ε-neighborhood. Point B has only 2, but lies inside A's ε-neighborhood. Point C has 1 and is not within ε of any core point. Classify A, B, C.

Q8 · STYLE 2: FIGURE READING

In the dendrogram shown, how many clusters does the dashed threshold produce, and what happens if you raise the threshold above all links?



A. 2 clusters; raising gives 8

B. 4 clusters; raising gives 1

C. 3 clusters; raising gives 2

D. 8 clusters; raising gives 4

Answer key — Quiz 3

Q1 — model answer. Not broken. Each run converges because neither step can increase J (assign to nearest centroid; centroid = mean) and there are finitely many assignments — but it converges to a LOCAL optimum that depends on initialization. Keep the run with lower J (389) or use more restarts.
Look up: W7 p11, p17, p25 · Reference PDF p. 79

Q2 → A. Assign: 1, 2 → c_1 (2 is 1 away from c_1 vs 4 from c_2); 6, 8, 9 → c_2. Update: $c_1 = (1+2)/2 = 1.5$; $c_2 = (6+8+9)/3 = 23/3 \approx 7.67$. J drops from 14 to about 5.2.
Why the others are wrong — **B**: That skips the update step. **C**: Put 6 in cluster 1 — but 6 is exactly at c_2. **D**: Left 6 out of cluster 2.
Look up: W7 p11–p12 · Reference PDF p. 83

Q3 → B. The curve levels off after component 3 → keep the components to its left: 2. Share = $(2.9 + 1.7)/6 = 4.6/6 \approx 77\%$ (eigenvalue ÷ #features).
Why the others are wrong — **A**: The drop from 1 to 2 is still steep — component 2 still carries a lot. **C**: Components 3–6 are on the flat part (little variance, mostly noise). **D**: PCA's purpose is to use fewer factors.
Look up: W7 p41–p42 · Reference PDF p. 81

Q4 → B. DBSCAN groups dense connected regions of arbitrary shape and leaves sparse points as outliers. K-Means assumes compact convex blobs and forces every point into a cluster.
Why the others are wrong — **A**: K-Means cannot follow non-convex shapes and is pulled by outliers. **C**: PCA reduces features; it does not cluster. **D**: Changing k does not fix the shape problem.
Look up: W7 p25, p33–p34 · Reference PDF p. 37

Q5 → B. Eigenvector = feature coefficients/direction of the component. Eigenvalue = explained variability.
Why the others are wrong — **A**: That is the eigenvalue. **C**: PCA does not count clusters. **D**: No p-values in PCA.
Look up: W7 p41 · Reference PDF p. 38

Q6 → C. Big drops until k = 3 (620 → 180), then only small gains (180 → 150 → 135). The bend (elbow) is at k = 3.
Why the others are wrong — **A**: k = 1 has the highest J. **B**: The drop from 2 to 3 is still huge (440). **D**: J always falls as k grows; the smallest J is not the goal.
Look up: W7 p19–p23 · Reference PDF p. 36

Q7 — model answer. A = core point (≥ min_samples in its ε-neighborhood). B = border point (in a core point's neighborhood but not core itself). C = outlier (neither core nor border).
Look up: W7 p35–p36 · Reference PDF p. 37

Q8 → B. Count the vertical lines the threshold crosses: {A,B}, {C,D}, {E,F}, {G,H} → 4. Above the top link, everything is merged into 1 cluster.
Why the others are wrong — **A**: Count crossings, not merges. **C**: There are 4 crossings at that height. **D**: 8 = all leaves (a threshold at height 0).
Look up: W7 p31–p32 · Reference PDF p. 37

Weak spots to review (tick the questions you missed)

Missed?	Q	Topic	Re-read
☐	Q1	K-Means convergence / local optimum	W7 p11, p17, p25 · PDF p. 79
☐	Q2	One K-Means step	W7 p11–p12 · PDF p. 83
☐	Q3	Reading a scree plot	W7 p41–p42 · PDF p. 81
☐	Q4	K-Means vs DBSCAN	W7 p25, p33–p34 · PDF p. 37
☐	Q5	Eigenvector vs eigenvalue in PCA	W7 p41 · PDF p. 38
☐	Q6	Elbow method	W7 p19–p23 · PDF p. 36
☐	Q7	DBSCAN core / border / outlier	W7 p35–p36 · PDF p. 37
☐	Q8	Reading a dendrogram	W7 p31–p32 · PDF p. 37

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Quiz 4. W5 — Model Performance Evaluation

Answer all questions first. Answer key on the next pages.

Q1 · STYLE 5: LIGHT MATH

Confusion matrix: TP = 30, FP = 10, FN = 20, TN = 140. Compute accuracy, precision, recall, and F1.

Q2 · STYLE 3: COMPARE & JUDGE

An email filter moves suspected spam out of the inbox. Losing a real email (a false positive) is very costly. Which metric should you prioritize?

A. Recall

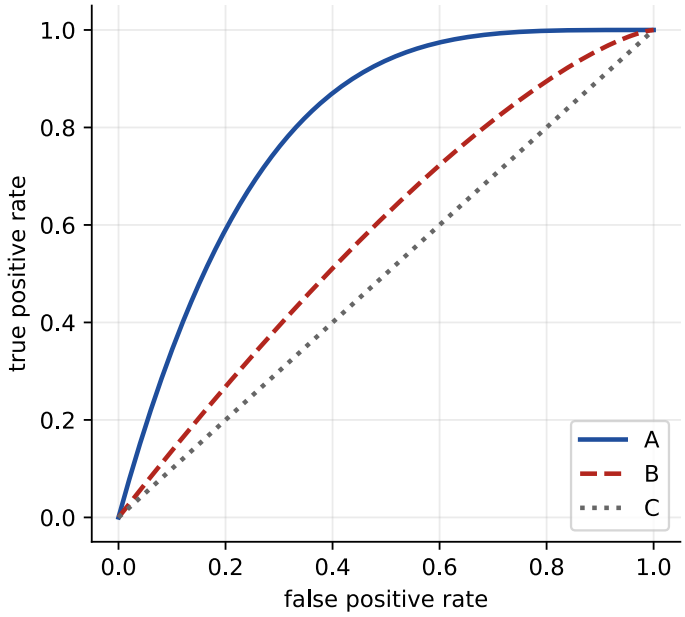
B. Precision

C. Accuracy

D. MAE

Q3 · STYLE 2: FIGURE READING

ROC curves for three models A, B, C are shown. Which statement is correct?



A. C is best because it is a straight line.

B. A is best (largest AUC); C is random guessing with AUC 0.5.

C. B is best because it is in the middle.

D. All have the same AUC.

Q4 · STYLE 1: WHAT / WHY

True class $y = 1$. Model A predicts $\hat{p} = 0.9$, model B predicts $\hat{p} = 0.6$. Both classify correctly at threshold 0.5. Which has the lower log loss, and what does accuracy miss?

A. B; accuracy misses nothing

B. A ($-\ln 0.9 \approx 0.11$ vs $-\ln 0.6 \approx 0.51$); accuracy ignores confidence

C. They are equal because both are correct

D. A; but log loss is higher for confident predictions

Q5 · STYLE 4: MECHANISM (HOW)

Why must hyperparameters (e.g. polynomial degree, α) be chosen on validation data or by cross-validation and never on the test set?

Q6 · STYLE 3: COMPARE & JUDGE

Switching from 5-fold to 20-fold cross-validation mainly causes:

A. Less computation and more bias

B. Smaller validation folds, more variability in the estimate, and more computation

C. No change at all

D. Larger validation folds

Q7 · STYLE 1: WHAT / WHY

Dart analogy: all the shots land in a tight group, but far from the bullseye. This model has:

A. Low bias, low variance

B. High bias, low variance

C. Low bias, high variance

D. High bias, high variance

Q8 · STYLE 5: LIGHT MATH

Prediction errors are $2, -2, 2, -6$. Compute MAE and RMSE. Which reacts more to the -6?

A. MAE $= 3$, RMSE ≈ 3.46; RMSE reacts more

B. MAE $= 3$, RMSE $= 3$; equal

C. MAE $= -1$, RMSE $= 12$

D. MAE $= 12$, RMSE ≈ 3.46; MAE reacts more

Answer key — Quiz 4

Q1 — model answer. Accuracy $= (30 + 140)/200 = 0.85$. Precision $= 30/(30 + 10) = 0.75$. Recall $= 30/(30 + 20) = 0.60$. F1 $= 2 \cdot 0.75 \cdot 0.60/(0.75 + 0.60) = 0.90/1.35 \approx 0.667$.
Look up: W5 p20 (F1: review file) · Reference PDF p. 83

Q2 → B. Precision $= \mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$ is hurt directly by false positives: 'when I say spam, am I right?'
Why the others are wrong — **A**: Recall penalizes misses (FN), i.e. spam reaching the inbox — less costly here. **C**: Accuracy hides the FP/FN balance, especially with imbalanced classes. **D**: MAE is a regression loss.
Look up: W5 p20–p21 · Reference PDF p. 28

Q3 → B. The curve closest to the top-left corner has the largest area under it. The diagonal is a random classifier (AUC $= 0.5$).
Why the others are wrong — **A**: The diagonal = TPR equals FPR at every cutoff = random. **C**: B lies below A everywhere → smaller AUC. **D**: The areas clearly differ.
Look up: W5 p22 · Reference PDF p. 27

Q4 → B. Log loss $= -\log$(probability given to the true class). Higher probability on the truth → smaller loss. Accuracy only sees the thresholded label, so it scores both the same.
Why the others are wrong — **A**: B gave the truth less probability → larger loss. **C**: Log loss uses the probability, not just correctness. **D**: Confident CORRECT predictions get LOW log loss; confident WRONG ones get very high loss.
Look up: W5 p18 · Reference PDF p. 26

Q5 — model answer. The test set must stay unseen so it gives an honest estimate of performance on new data. If you tune on it, the test set is effectively used for training (data leakage), and the reported test error becomes optimistic.
Look up: W5 p8–p9; W4 p30 · Reference PDF p. 26

Q6 → B. Larger k → smaller validation sets (more variability) and more models trained on larger folds (more computation). Smaller k → more bias.
Why the others are wrong — **A**: Computation goes UP with k; more bias belongs to small k. **C**: Both variance and cost change. **D**: Each fold is $1/k$ of the data — smaller.
Look up: W5 p24 · Reference PDF p. 27

Q7 → B. Bias = average distance from the bullseye (far → high). Variance = distance of shots from one another (tight → low).
Why the others are wrong — **A**: It would be on the bullseye. **C**: That would be scattered shots centred on the bullseye. **D**: Shots would be both scattered and off-center.
Look up: W5 p5 · Reference PDF p. 26

Q8 → A. MAE $= (2+2+2+6)/4 = 3$. MSE $= (4+4+4+36)/4 = 12 \rightarrow$ RMSE $= \sqrt{12} \approx 3.46$. Squaring makes the big error dominate, so RMSE is more outlier-sensitive.
Why the others are wrong — **B**: Squaring changes the result. **C**: MAE uses absolute values (never negative); 12 is MSE, not RMSE. **D**: 12 is the sum of |errors|, not the mean.
Look up: W5 p13–p14 · Reference PDF p. 27

Weak spots to review (tick the questions you missed)

Missed?	Q	Topic	Re-read
☐	Q1	Confusion-matrix metrics	W5 p20 (F1: review file) · PDF p. 83
☐	Q2	Precision vs recall choice	W5 p20–p21 · PDF p. 28
☐	Q3	Reading ROC / AUC	W5 p22 · PDF p. 27
☐	Q4	Log loss vs accuracy	W5 p18 · PDF p. 26
☐	Q5	Validation vs test set	W5 p8–p9; W4 p30 · PDF p. 26
☐	Q6	Choice of k in k-fold CV	W5 p24 · PDF p. 27
☐	Q7	Bias vs variance	W5 p5 · PDF p. 26
☐	Q8	MAE vs RMSE	W5 p13–p14 · PDF p. 27

*Summary line: if you missed ≥ 3 here, re-read this week's **Mechanisms** and **Comparisons** blocks before the next quiz.*

Quiz 5. W12 — Intro to NLP (incl. LDA)

Answer all questions first. Answer key on the next pages.

Q1 · STYLE 1: WHAT / WHY

In Latent Dirichlet Allocation, a "topic" is best described as:

A. A cluster of documents

B. A probability distribution over words

C. A single keyword chosen by a human

D. A labeled class for supervised learning

Q2 · STYLE 3: COMPARE & JUDGE

You increase LDA's α (document–topic density) and keep β fixed. What changes?

A. Each topic uses more words

B. Each document is made of more topics

C. The number of topics k doubles

D. Documents become single-topic

Q3 · STYLE 4: MECHANISM (HOW)

In the LDA update, a word w in document d is being reassigned. Name the two probabilities the algorithm multiplies, and say in plain words what each one asks.

Q4 · STYLE 3: COMPARE & JUDGE

Which normalization maps "was" $\rightarrow$ "be" and "better" $\rightarrow$ "good"?

A. Stemming

B. Lemmatization

C. Tokenization

D. TF-IDF

Q5 · STYLE 1: WHAT / WHY

Why does a Transformer train faster than an RNN on long sentences?

A. It has no parameters

B. It processes all words at once with self-attention instead of one at a time

C. It ignores word order completely

D. It uses n-grams

Q6 · STYLE 4: MECHANISM (HOW)

In an LSTM, which gate removes information that is no longer important?

A. Input gate

B. Output gate

C. Forget gate

D. Attention gate

Q7 · STYLE 5: LIGHT MATH

Bigram model. In a corpus, "the" occurs 8 times; "the cat" occurs 3 times and "the dog" once. What is P(cat | the)?

A. 3/4

B. 3/8

C. 1/8

D. 3/11

Q8 · STYLE 3: COMPARE & JUDGE

Which pairing of model and task is correct?

A. GPT (decoder-only, left-to-right) → text generation; BERT (encoder-only, bidirectional) → sentiment / NER

B. BERT → text generation; GPT → named-entity recognition

C. BART is decoder-only

D. GPT is pre-trained with masked language modeling

Answer key — Quiz 5

Q1 → B. Topic = distribution over words; each document = distribution (mixture) over topics. Topics are latent — inferred from observed words.
Why the others are wrong — **A**: Documents are MIXTURES of topics, not members of one cluster. **C**: Topics are learned, not hand-picked. **D**: LDA is unsupervised — no labels.
Look up: W12 p11 · Reference PDF p. 61

Q2 → B. Higher α → documents are made of more topics. Higher β → topics are made of more words.
Why the others are wrong — **A**: That is the effect of a higher β. **C**: k is chosen separately (step 1 of the algorithm). **D**: That would be a LOWER α.
Look up: W12 p14 · Reference PDF p. 62

Q3 — model answer. P(t | d) — 'does this document like topic t?' (share of d's other words currently in t). P(w | t) — 'does topic t like this word?' (how often w appears in t in the other documents). Update: P(w | t) = P(t | d) · P(w | t), repeated until the distributions converge.
Look up: W12 p13 · Reference PDF p. 62

Q4 → B. Lemmatization uses linguistics to get the root word, so it handles irregular forms. Stemming only chops prefixes/suffixes from a list.
Why the others are wrong — **A**: Chopping suffixes cannot turn 'was' into 'be'. **C**: Tokenization only splits text into words. **D**: TF-IDF weights words; it does not normalize them.
Look up: W12 p10 · Reference PDF p. 61

Q5 → B. Transformers eliminate sequential dependency → parallel processing; attention links words at any distance.
Why the others are wrong — **A**: Transformers have very many parameters. **C**: Position information is still used. **D**: N-grams are the traditional approach.
Look up: W12 p24, p29 · Reference PDF p. 64

Q6 → C. Forget gate removes unimportant info; input gate adds new important info; output gate decides what to send forward.
Why the others are wrong — **A**: The input gate ADDS information. **B**: The output gate chooses what to pass on. **D**: Attention belongs to Transformers, not LSTM gates.
Look up: W12 p28 · Reference PDF p. 64

Q7 → B. P(w_i | w_{i−1}) = count(the, cat)/count(the) = 3/8 (Markov assumption: depends only on the previous word).
Why the others are wrong — **A**: Divided by the cat+dog count (4), not by count(the). **C**: That is P(dog | the). **D**: Added counts that do not belong together.
Look up: W12 p23 (formula: standard) · Reference PDF p. 63

Q8 → A. Deck table: BERT = encoder, bidirectional, MLM, sentiment/NER; GPT = decoder, unidirectional, autoregressive, generation; BART = encoder–decoder, span corruption, translation/summarization.
Why the others are wrong — **B**: Swapped. **C**: BART is encoder–decoder. **D**: MLM is BERT's objective; GPT is autoregressive.
Look up: W12 p19 · Reference PDF p. 62

Weak spots to review (tick the questions you missed)

Missed?	Q	Topic	Re-read
☐	Q1	LDA: what a topic is	W12 p11 · PDF p. 61
☐	Q2	LDA α vs β	W12 p14 · PDF p. 62
☐	Q3	LDA algorithm steps	W12 p13 · PDF p. 62
☐	Q4	Stemming vs lemmatization	W12 p10 · PDF p. 61
☐	Q5	Transformer vs RNN	W12 p24, p29 · PDF p. 64
☐	Q6	LSTM gates	W12 p28 · PDF p. 64
☐	Q7	Bigram probability	W12 p23 (formula: standard) · PDF p. 63
☐	Q8	BERT vs GPT vs BART	W12 p19 · PDF p. 62

*Summary line: if you missed ≥ 3 here, re-read this week's **Mechanisms** and **Comparisons** blocks before the next quiz.*

Quiz 6. W11 — Deep Learning

Answer all questions first. Answer key on the next pages.

Q1 · STYLE 5: LIGHT MATH

Perceptron with unit step (output 1 if $z \geq 0$). $w = (0.5, -1)$, $b = 0$, learning rate 0.2. Input $x = (2, 1)$, true class 0. Do one update with the deck rule weight = weight + rate × error × input (error = actual − predicted). Give the new w, b and the new prediction.

Q2 · STYLE 1: WHAT / WHY

A slide writes the backprop update as weight = weight + $r \cdot \partial L/\partial w$. If $\partial L/\partial w = +4$ at the current weight, what would that update do to the loss?

A. Decrease it — that is the correct rule

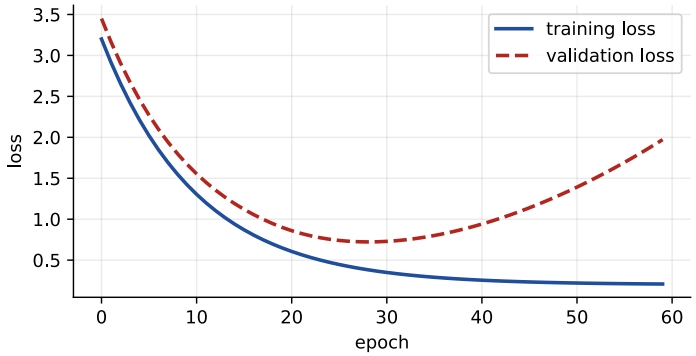
B. Increase it — it moves uphill; the correct rule uses a minus sign

C. Nothing, because r is small

D. It sets the weight to 4

Q3 · STYLE 2: FIGURE READING

The loss curves show training loss still falling while validation loss starts rising after about epoch 20. Diagnosis and a deck-supported remedy?



A. Underfitting — add more epochs

B. Overfitting — stop training earlier / use dropout or regularization

C. Perfect fit — keep training

D. Learning rate too small — the loss is stuck

Q4 · STYLE 4: MECHANISM (HOW)

Explain why a deep network with sigmoid activations suffers from vanishing gradients. Use the maximum value of the sigmoid derivative.

Q5 · STYLE 5: LIGHT MATH

1,000,000 training samples, mini-batch size 1,000. How many weight updates per epoch for full-batch GD, mini-batch GD, and SGD?

A. 1 / 1,000 / 1,000,000

B. 1,000 / 1 / 1,000,000

C. 1,000,000 / 1,000 / 1

D. 1 / 1,000,000 / 1,000

Q6 · STYLE 3: COMPARE & JUDGE

Which of these models is GENERATIVE according to the deck?

A. Support vector machine

B. Naive Bayes classifier

C. A decision boundary learned by a perceptron

D. k-NN

Q7 · STYLE 5: LIGHT MATH

Chain rule: $\hat{y} = w \cdot z$, $L = (y - \hat{y})^2$. With $y = 5$, $w = 1$, $z = 2$ and learning rate 0.05, what is $\partial L / \partial w$ and the updated w?

A. $\partial L / \partial w = -12$, new $w = 1.6$

B. $\partial L / \partial w = 12$, new $w = 1.6$

C. $\partial L / \partial w = -6$, new $w = 1.3$

D. $\partial L / \partial w = -12$, new $w = 0.4$

Q8 · STYLE 1: WHAT / WHY

Why does a multilayer network need NON-linear activation functions?

A. To make training faster

B. Because stacking linear layers is still one linear function, which can only draw linear boundaries

C. Because outputs must be between 0 and 1

D. To avoid using a loss function

Answer key — Quiz 6

Q1 — model answer. $z = 0.5 \cdot 2 + (-1) \cdot 1 + 0 = 0 \rightarrow$ predicted 1. Error $= 0 - 1 = -1$. $w_1 = 0.5 + 0.2(-1)(2) = 0.1$; $w_2 = -1 + 0.2(-1)(1) = -1.2$; $b = 0 + 0.2(-1)(1) = -0.2$. New $z = 0.2 - 1.2 - 0.2 = -1.2 < 0 \rightarrow$ predicts 0 (now correct).
Look up: W11 p7 · Reference PDF p. 85

Q2 → B. The gradient points uphill. Adding it increases the loss. Standard: $w \leftarrow w - r \cdot \partial L / \partial w$ (deck W11 p18 and W4 p15 use minus; W11 p11's plus is likely a typo).
Why the others are wrong — **A**: Moving with a positive slope raises L. **C**: Any step uphill increases the loss. **D**: The update is a step, not an assignment to the gradient.
Look up: W11 p11 vs p18 · Reference PDF p. 58

Q3 → B. Validation loss turning up while training loss decreases = memorizing the training data (overfitting). Dropout is the deck's 'improving training performance' tool; stopping near the validation minimum also helps.
Why the others are wrong — **A**: Underfitting = both losses high. **C**: The gap is growing. **D**: Training loss is still decreasing well.
Look up: W11 p13–p14, p17 · Reference PDF p. 56

Q4 — model answer. Backprop multiplies one activation derivative per layer (chain rule). The sigmoid derivative is at most 0.25, so after n layers the gradient is scaled by at most 0.25^n (5 layers ≈ 0.001). Early layers get almost no update and stop learning. Fixes: ReLU, batch normalization; for sequences, LSTM.
Look up: W11 p20–p22 · Reference PDF p. 56

Q5 → A. Full batch: one update after all n samples. Mini-batch: b = n / batch size = 1,000 updates. SGD: one update per sample = n.
Why the others are wrong — **B**: Full batch makes only ONE update per epoch. **C**: Reversed. **D**: Mini-batch and SGD swapped.
Look up: W11 p23 · Reference PDF p. 56

Q6 → B. Generative models estimate probability distributions of the data (can generate new data): some unsupervised models, naive Bayes, LLMs. SVM is the deck's discriminative example.
Why the others are wrong — **A**: The deck's example of a DISCRIMINATIVE model. **C**: A boundary = discriminative. **D**: kNN models no distribution.
Look up: W11 p24 · Reference PDF p. 57

Q7 → A. $\hat{y} = 2$. $\partial L/\partial w = f'(\hat{y}) \cdot g'(w) = -2(y - \hat{y}) \cdot z = -2(3)(2) = -12$. Update: $w = 1 - 0.05(-12) = 1.6$ (w goes up, $\hat{y}$ moves toward 5).
Why the others are wrong — **B**: Sign of the gradient is wrong (prediction too LOW → negative gradient). **C**: Forgot the factor $z = 2$. **D**: Used a plus-sign update (moved the wrong way).
Look up: W11 p21 · Reference PDF p. 58

Q8 → B. A perceptron is good for linear boundaries only; complex data needs multilayer networks with non-linear activation.
Why the others are wrong — **A**: Speed is not the reason. **C**: ReLU outputs are not bounded by 1. **D**: Every network still needs a loss.
Look up: W11 p8 · Reference PDF p. 55

Weak spots to review (tick the questions you missed)

Missed?	Q	Topic	Re-read
☐	Q1	Perceptron update	W11 p7 · PDF p. 85
☐	Q2	Gradient update sign	W11 p11 vs p18 · PDF p. 58
☐	Q3	Reading loss curves	W11 p13–p14, p17 · PDF p. 56
☐	Q4	Vanishing gradients	W11 p20–p22 · PDF p. 56
☐	Q5	Mini-batch vs stochastic GD	W11 p23 · PDF p. 56
☐	Q6	Generative vs discriminative	W11 p24 · PDF p. 57
☐	Q7	Chain rule gradient	W11 p21 · PDF p. 58
☐	Q8	Why non-linear activations	W11 p8 · PDF p. 55

*Summary line: if you missed ≥ 3 here, re-read this week's **Mechanisms** and **Comparisons** blocks before the next quiz.*

Quiz 7. W8 — Decision Trees and Random Forests

Answer all questions first. Answer key on the next pages.

Q1 · STYLE 5: LIGHT MATH

A node has 6 class-A and 2 class-B instances. A split sends 4 A to the left and 2 A + 2 B to the right. What is the weighted (overall) Gini index after the split?

A. 0.375

B. 0.25

C. 0.5

D. 0

Q2 · STYLE 5: LIGHT MATH

Same split as above. Using entropy, compute the information gain. (Use $H(¼) \approx 0.811$ and $H(½) = 1$.)

Q3 · STYLE 1: WHAT / WHY

What is the maximum possible Gini index for a node with 3 equally likely classes?

A. 1

B. 2/3

C. 1/3

D. 0.5

Q4 · STYLE 1: WHAT / WHY

Why does a random forest usually generalize better than one deep decision tree?

A. Each tree uses all data and all features, so they agree more

B. Bootstrap samples + random feature subsets ($m < M$) decorrelate the trees, and voting/averaging reduces variance

C. Random forests never split on numeric features

D. Random forests are easier to interpret

Q5 · STYLE 4: MECHANISM (HOW)

A regression tree is trained on houses priced \$100k–\$500k. You ask it to price a mansion much larger than any training house. What will it predict and why?

Q6 · STYLE 3: COMPARE & JUDGE

Compared with linear regression, which is TRUE of a regression tree according to the deck?

A. It is heavily affected by outliers

B. Missing values can be treated as a separate class and it is not affected by outliers

C. It makes the statistical significance of each feature easy to measure

D. It fits linear data better

Q7 · STYLE 4: MECHANISM (HOW)

Which is a stopping criterion used to keep CART from overfitting?

A. Split until every leaf has exactly one instance

B. Set a minimum number of instances per leaf (or a maximum number of terminal nodes)

C. Always split on the feature with the highest Gini

D. Standardize all features

Answer key — Quiz 7

Q1 → B. Left: pure → 0. Right: $1 - 0.5^2 - 0.5^2 = 0.5$. Weighted: $(4/8)(0) + (4/8)(0.5) = 0.25$. (Parent was $1 - 0.75^2 - 0.25^2 = 0.375$, so the split helps.)
Why the others are wrong — **A**: That is the PARENT Gini. **C**: That is only the right child, unweighted. **D**: Only the left child is pure.
Look up: W8 p16–p17 · Reference PDF p. 42

Q2 — model answer. Parent $H(6/8, 2/8) = H(¼) \approx 0.811$. Children: left 0, right 1. Weighted = ½·0 + ½·1 = 0.5. IG = $0.811 - 0.5 \approx 0.311$ bits. (Formula: not in deck — standard.)
Look up: W8 p16 (entropy: standard) · Reference PDF p. 84

Q3 → B. $1 - 3 \cdot (1/3)^2 = 1 - 1/3 = 2/3$. In general the max is $1 - 1/k$ (0.5 for two classes) — the deck's '0 to 1' is an approximation.
Why the others are wrong — **A**: 1 is only approached with very many classes. **C**: That is Σp^2 , not $1 - \Sigma p^2$. **D**: That is the two-class maximum.
Look up: W8 p16 · Reference PDF p. 42

Q4 → B. Double randomization reduces correlation among trees; combining many decorrelated trees cancels individual errors.
Why the others are wrong — **A**: Using the same data and features would make the trees nearly identical (no gain). **C**: They split on any feature type. **D**: A forest is LESS interpretable than one tree.
Look up: W8 p20 · Reference PDF p. 42

Q5 — model answer. At most the mean of the leaf the mansion falls into (around the highest training prices). Each leaf predicts the mean of its region, so trees do not extrapolate beyond the training range.
Look up: W8 p5, p13 · Reference PDF p. 43

Q6 → B. Deck table: trees handle missing values as a separate class and are not affected by outliers; linear regression fits linear data better and gives significance easily.
Why the others are wrong — **A**: That is linear regression. **C**: That is linear regression. **D**: That is linear regression.
Look up: W8 p6 · Reference PDF p. 41

Q7 → B. Stopping criteria limit tree size: minimum instances in a leaf, or maximum terminal nodes.
Why the others are wrong — **A**: That is maximal overfitting. **C**: CART picks the LOWEST Gini. **D**: Trees do not need scaling.
Look up: W8 p12, p19 · Reference PDF p. 41

Weak spots to review (tick the questions you missed)

Missed?	Q	Topic	Re-read
☐	Q1	Gini index calculation	W8 p16–p17 · PDF p. 42
☐	Q2	Entropy / information gain	W8 p16 (entropy: standard) · PDF p. 84
☐	Q3	Gini range	W8 p16 · PDF p. 42
☐	Q4	Random forest	W8 p20 · PDF p. 42
☐	Q5	Trees don't extrapolate	W8 p5, p13 · PDF p. 43
☐	Q6	Regression tree vs linear regression	W8 p6 · PDF p. 41
☐	Q7	Stopping criteria	W8 p12, p19 · PDF p. 41

*Summary line: if you missed ≥ 3 here, re-read this week's **Mechanisms** and **Comparisons** blocks before the next quiz.*

Quiz 8. W9 — Ensemble Learning

Answer all questions first. Answer key on the next pages.

Q1 · STYLE 3: COMPARE & JUDGE

Which ensemble method CANNOT fit its base models in parallel, and why?

A. Bagging, because of bootstrapping

B. Boosting, because each base model needs the instance weights updated by the previous one

C. Stacking, because of the meta model

D. Random forest, because of random features

Q2 · STYLE 5: LIGHT MATH

AdaBoost: 10 instances, equal weights 1/10. The first base model misclassifies 2 of them. Using the deck's rule (ε = fraction wrong; multiply misclassified weights by $\sqrt{((1-\varepsilon)/\varepsilon)}$, divide correct ones by it, then normalize), give the new normalized weight of a misclassified and of a correct instance.

Q3 · STYLE 5: LIGHT MATH

Bootstrap sample drawn from instances {1, 2, 3, 4, 5, 6}: 2, 2, 5, 1, 5, 6. Which instances are out-of-bag, and what are they used for?

A. {3, 4} — to estimate the base model's error like a validation set

B. {2, 5} — they are the most important

C. {1, 6} — they appear once

D. None — bootstrap uses every instance

Q4 · STYLE 1: WHAT / WHY

In stacking for a CLASSIFICATION task, which meta model does the deck use?

A. Linear regression

B. Logistic regression

C. k-means

D. Another bootstrap sample

Q5 · STYLE 5: LIGHT MATH

Five base classifiers vote Yes, No, Yes, Yes, No; four base regressors predict 12, 15, 9, 20. What are the ensemble outputs?

A. Yes (60%) and 14

B. No and 14

C. Yes and 20

D. Yes and 56

Q6 · STYLE 3: COMPARE & JUDGE

What is the key difference between plain bagging of decision trees and a random forest?

A. Random forests do not use bootstrap samples

B. Random forests also restrict each split to a random subset of features ($m < M$)

C. Bagging updates instance weights

D. Random forests use a meta model

Q7 · STYLE 4: MECHANISM (HOW)

When does AdaBoost stop adding base models?

A. After exactly 10 models

B. When the maximum number of base models is reached or the error no longer changes

C. When every instance has equal weight

D. Never — it runs forever

Answer key — Quiz 8

Q1 → B. Boosting is sequential: it updates instance weights after each base model to emphasize the poorly predicted instances.

Why the others are wrong — **A**: Bootstrap samples are independent → parallel. **C**: Stacking's base models are fitted independently; only the meta model comes after. **D**: Trees are independent → parallel.

Look up: W9 p4–p5, p7 · Reference PDF p. 45

Q2 — model answer. $\varepsilon = 2/10 = 0.2 \rightarrow$ factor $\sqrt{(0.8/0.2)} = \sqrt{4} = 2$. Relative weights: misclassified 2 each, correct ½ each. Sum $= 2{\cdot}2 + 8{\cdot}½ = 8$. Normalized: misclassified $2/8 = 0.25$ each; correct $0.5/8 = 0.0625$ each (check: $0.5 + 0.5 = 1$).

Look up: W9 p10–p11 · Reference PDF p. 46

Q3 → A. OOB = never drawn = $\{3, 4\}$. They were not used to train this base model, so predicting them is an honest error estimate.

Why the others are wrong — **B**: Those were drawn twice — they are in the bag. **C**: Drawn once still means in the bag. **D**: Sampling with replacement leaves some out.

Look up: W9 p16 · Reference PDF p. 45

Q4 → B. Classification → logistic regression meta model; regression → linear regression meta model.

Why the others are wrong — **A**: That is for regression tasks. **C**: k-means is unsupervised clustering. **D**: Stacking trains base models on the entire dataset, not bootstraps.

Look up: W9 p18 · Reference PDF p. 46

Q5 → A. Classification: majority vote (3 of 5 = 60% Yes). Regression: average $(12 + 15 + 9 + 20)/4 = 56/4 = 14$.

Why the others are wrong — **B**: Only 2 of 5 voted No. **C**: Ensembles average; they do not take the max. **D**: 56 is the sum, not the average.

Look up: W9 p6 · Reference PDF p. 45

Q6 → B. Random forest = bagging + random feature subsets, which decorrelates the trees further.

Why the others are wrong — **A**: Both use bootstrap samples. **C**: Weight updates are boosting. **D**: A meta model is stacking.

Look up: W8 p20; W9 p12–p15 · Reference PDF p. 42

Q7 → B. AdaBoost has no fixed number of base models; it stops at the max or when the error stops changing (often early). In the deck's hepatitis example it stopped once model 2 got everything right.
Why the others are wrong — **A**: No fixed number. **C**: Weights become UNequal as it runs. **D**: It has stopping rules.
Look up: W9 p8–p10 · Reference PDF p. 45

Weak spots to review (tick the questions you missed)

Missed?	Q	Topic	Re-read
☐	Q1	Sequential vs parallel ensembles	W9 p4–p5, p7 · PDF p. 45
☐	Q2	AdaBoost re-weighting	W9 p10–p11 · PDF p. 46
☐	Q3	Out-of-bag instances	W9 p16 · PDF p. 45
☐	Q4	Stacking meta model	W9 p18 · PDF p. 46
☐	Q5	Combining predictions	W9 p6 · PDF p. 45
☐	Q6	Bagging vs random forest	W8 p20; W9 p12–p15 · PDF p. 42
☐	Q7	AdaBoost stopping	W9 p8–p10 · PDF p. 45

*Summary line: if you missed ≥ 3 here, re-read this week's **Mechanisms** and **Comparisons** blocks before the next quiz.*

Quiz 9. W10 — Recommender Systems

Answer all questions first. Answer key on the next pages.

Q1 · STYLE 3: COMPARE & JUDGE

A brand-new user signs up and has no interaction history. What is this problem called, and what is the deck's suggested solution?

A. Serendipity — show random items

B. Cold start — query the user for more information (e.g. ask preferences, import contacts)

C. Scalability — use matrix factorization

D. Feature extraction — use neural networks

Q2 · STYLE 5: LIGHT MATH

User vector $u = (1, 0, 1)$. Item $A = (1, 0, 1)$, item $B = (1, 1, 1)$. Compare dot product and cosine similarity.

A. Dot: $A = 2$, $B = 2$ (tie); cosine: $A = 1$, $B \approx 0.82 \rightarrow$ cosine prefers A

B. Dot: $A = 2$, $B = 3$; cosine prefers B

C. Both measures prefer B

D. Dot: $A = 1$, $B = 1$

Q3 · STYLE 5: LIGHT MATH

$A = \{x, y, z, w\}$, $B = \{y, z, v\}$. Compute Jaccard and Dice similarity.

A. Jaccard $= 0.4$, Dice ≈ 0.57

B. Jaccard $= 0.5$, Dice $= 0.4$

C. Jaccard $= 2$, Dice $= 4$

D. Jaccard ≈ 0.57, Dice $= 0.4$

Q4 · STYLE 5: LIGHT MATH

There are $N = 10$ documents. The word "the" appears in all 10; the word "cold-start" appears in 1. Using log base 10, what are their IDF values?

A. the: 1, cold-start: 0

B. the: 0, cold-start: 1

C. both 10

D. the: 10, cold-start: 1

Q5 · STYLE 1: WHAT / WHY

Word2Vec skip-gram is trained to:

A. Predict the center word from its context words

B. Predict the context words from the center word

C. Factor the global co-occurrence matrix

D. Count words in a document

Q6 · STYLE 3: COMPARE & JUDGE

A website helps people buy a car — purchases are rare and cars have many complex features. Which recommender type fits best?

A. Collaborative filtering

B. Knowledge-based

C. Pure content-based on viewing history

D. Bag of words

Q7 · STYLE 4: MECHANISM (HOW)

In collaborative filtering with Pearson similarity, why do we subtract each user's mean rating, and why only use items both users rated?

Q8 · STYLE 1: WHAT / WHY

In matrix factorization for recommendation, how is a missing rating M_ij predicted?

A. As the average of all ratings

B. As the dot product of user i's vector and item j's vector

C. As the cosine of the two vectors' angle only

D. As the number of users who rated item j

Answer key — Quiz 9

Q1 → B. Cold start = no informative history (new user, or naturally sparse interactions). Solution: ask the user.

Why the others are wrong — **A**: Serendipity = recommendations become repetitive. **C**: Scalability = millions of users/items. **D**: Feature extraction = too many items to label by hand.

Look up: W10 p14 · Reference PDF p. 49

Q2 → A. Dot products: 1+0+1 = 2 for both. Cosine = dot/(|u||item|): A = $2/(\sqrt{2}\cdot\sqrt{2}) = 1$; B = $2/(\sqrt{2}\cdot\sqrt{3}) \approx 0.82$. Cosine penalizes the 'un-specific' item B.

Why the others are wrong — **B**: u has a 0 in position 2, so B's extra feature adds nothing to the dot product. **C**: Cosine divides by B's larger length. **D**: Count the matching 1s: there are 2.

Look up: W10 p18–p19 · Reference PDF p. 50

Q3 → A. $|A \cap B| = 2$ ({y, z}); $|A \cup B| = 5$. Jaccard = $2/5 = 0.4$. Dice = $2\cdot2/(4+3) = 4/7 \approx 0.57$.

Why the others are wrong — **B**: Union has 5 elements, not 4. **C**: These are counts, not ratios. **D**: Swapped (Dice ≥ Jaccard always).

Look up: W10 p45–p46 · Reference PDF p. 51

Q4 → B. idf = log(N / #docs containing t): the → $\log(10/10) = \log 1 = 0$; cold-start → $\log(10/1) = 1$. Rare words get more weight; stop words get zero.

Why the others are wrong — **A**: Reversed. **C**: IDF uses a ratio and a log. **D**: Forgot the log and the ratio.

Look up: W10 p26 · Reference PDF p. 51

Q5 → B. Skip-gram: center → context, P(w in context | center). CBOW is the reverse; GloVe factors co-occurrence; BoW counts.

Why the others are wrong — **A**: That is CBOW. **C**: That is GloVe. **D**: That is bag of words.

Look up: W10 p31–p36 · Reference PDF p. 50

Q6 → B. Knowledge-based: for sparse interactions and complex item features; query the user's needs and apply domain knowledge (if user needs X, recommend Y).

Why the others are wrong — **A**: CF needs plenty of interaction history. **C**: Viewing history is too sparse/uninformative. **D**: BoW is a text representation, not a recommender type.

Look up: W10 p10–p11 · Reference PDF p. 48

Q7 — model answer. Subtracting the mean removes each user's rating bias (generous vs harsh raters), so we compare taste, not scale. The similarity needs complete vectors, but most ratings are missing, so we only use the co-rated items S_xy.

Look up: W10 p51, p61–p62 · Reference PDF p. 51

Q8 → B. Learn d-length vectors so ⟨user_i, item_j⟩ ≈ M_ij; user vector = preferences, item vector = features.
Why the others are wrong — **A**: That ignores who the user is. **C**: The dot product (not just the angle) is used. **D**: That is a popularity count.
Look up: W10 p9 · Reference PDF p. 49

Weak spots to review (tick the questions you missed)

Missed?	Q	Topic	Re-read
☐	Q1	Cold start	W10 p14 · PDF p. 49
☐	Q2	Dot product vs cosine	W10 p18–p19 · PDF p. 50
☐	Q3	Jaccard and Dice	W10 p45–p46 · PDF p. 51
☐	Q4	IDF calculation	W10 p26 · PDF p. 51
☐	Q5	Skip-gram vs CBOW	W10 p31–p36 · PDF p. 50
☐	Q6	Choosing the recommender type	W10 p10–p11 · PDF p. 48
☐	Q7	Pearson similarity in CF	W10 p51, p61–p62 · PDF p. 51
☐	Q8	Matrix factorization	W10 p9 · PDF p. 49

*Summary line: if you missed ≥ 3 here, re-read this week's **Mechanisms** and **Comparisons** blocks before the next quiz.*

Quiz 10. W2 — Math Review: Probability, Statistics, Linear Algebra

Answer all questions first. Answer key on the next pages.

Q1 · STYLE 5: LIGHT MATH

A disease affects 1% of people. A test detects it 90% of the time when present, and gives a false positive 10% of the time when absent. If a person tests positive, what is P(disease | positive)? (Illustrative numbers.)

Q2 · STYLE 1: WHAT / WHY

$P(A) = 0.3$ and $P(B) = 0.4$. If A and B are mutually exclusive, which is true?

A. They are independent, so $P(A \cap B) = 0.12$

B. $P(A \cap B) = 0$ and they are dependent

C. $P(A \cup B) = 0.12$

D. $P(A \mid B) = 0.3$

Q3 · STYLE 5: LIGHT MATH

A fair coin is flipped 4 times. What is the probability of exactly 2 heads, and what are $E[X]$ and $Var[X]$ for the number of heads?

A. $3/8$; $E = 2$; $Var = 1$

B. $1/2$; $E = 2$; $Var = 2$

C. $1/4$; $E = 2$; $Var = 1$

D. $3/8$; $E = 4$; $Var = 1$

Q4 · STYLE 1: WHAT / WHY

A distribution of house prices has positive skewness. What does that mean?

A. An extended left tail (many very cheap houses)

B. An extended right tail (a few very expensive houses)

C. Heavy tails on both sides

D. It is perfectly symmetric

Q5 · STYLE 5: LIGHT MATH

Data: 2, 4, 6. What are the population variance and the sample variance?

A. $8/3 \approx 2.67$ and 4

B. 4 and $8/3$

C. 8 and 4

D. 2 and 2

Q6 · STYLE 5: LIGHT MATH

$A = [[3, 0], [0, 1]]$. Which statement is correct?

A. $(1, 0)$ is an eigenvector with $\lambda = 3$, and $A \cdot (2, 1) = (6, 1)$

B. $(1, 1)$ is an eigenvector with $\lambda = 3$

C. A has no eigenvectors

D. $A \cdot (2, 1) = (3, 1)$

Q7 · STYLE 1: WHAT / WHY

Which matrix CANNOT be inverted?

A. $[[2, 0], [0, 5]]$

B. $[[1, 2], [2, 4]]$

C. $[[1, 0], [0, 1]]$

D. $[[0, 1], [1, 0]]$

Q8 · STYLE 1: WHAT / WHY

A hypothesis test gives p-value $= 0.01$. Which interpretation matches the deck?

A. There is a 1% chance the null hypothesis is true

B. If the null hypothesis were true, data this extreme (or more) would occur only 1% of the time

C. The effect size is 1%

D. The alternative hypothesis is 99% likely

Answer key — Quiz 10

Q1 — model answer. Prior 0.01; likelihood 0.9. Marginal $P(+) = 0.9 \cdot 0.01 + 0.1 \cdot 0.99 = 0.009 + 0.099 = 0.108$ (law of total probability). Posterior $= 0.009 / 0.108 = 1/12 \approx 0.083$ — only about 8%, because the disease is rare.
Look up: W2 p22, p25 · Reference PDF p. 16

Q2 → B. Mutually exclusive → cannot both happen → $P(A \cap B) = 0$. That makes them DEPENDENT (knowing B happened makes A impossible). If they were independent, $P(A \cap B)$ would be 0.12.
Why the others are wrong — **A**: Independence and mutual exclusivity are different. **C**: $P(A \cup B) = 0.3 + 0.4 - 0 = 0.7$. **D**: $P(A \mid B) = 0$ because A cannot happen when B does.
Look up: W2 p23–p24 · Reference PDF p. 14

Q3 → A. Binomial: $C(4,2) \cdot (½)^2 \cdot (½)^2 = 6/16 = 3/8$. $E = np = 2$; $Var = npq = 4 \cdot ½ \cdot ½ = 1$.
Why the others are wrong — **B**: Var = npq, not np. **C**: Forgot the $C(4,2) = 6$ orderings. **D**: $E = np = 2$, not n.
Look up: W2 p27 · Reference PDF p. 15

Q4 → B. Positive skewness = extended right tail; negative = extended left tail. Tail heaviness on both sides is kurtosis.
Why the others are wrong — **A**: That is negative skewness. **C**: That is high kurtosis. **D**: Symmetric means skewness ≈ 0.
Look up: W2 p7 · Reference PDF p. 13

Q5 → A. Mean = 4; squared deviations 4, 0, 4 → sum 8. Population: $8/n = 8/3$. Sample: $8/(n-1) = 8/2 = 4$.
Why the others are wrong — **B**: Swapped: the sample version divides by the smaller $n - 1$, so it is larger. **C**: 8 is the sum, not the variance. **D**: That is a standard deviation-like guess.
Look up: W2 p8 · Reference PDF p. 15

Q6 → A. $A \cdot (1, 0) = (3, 0) = 3 \cdot (1, 0) \rightarrow \lambda = 3$ (also $(0, 1)$ with $\lambda = 1$). Matrix–vector: row 1 · $(2, 1) = 6$; row 2 · $(2, 1) = 1$.
Why the others are wrong — **B**: $A \cdot (1, 1) = (3, 1)$, not a multiple of $(1, 1)$. **C**: Diagonal matrices have the axis directions as eigenvectors. **D**: Row 1 gives $3 \cdot 2 + 0 \cdot 1 = 6$.
Look up: W2 p37, p46 · Reference PDF p. 15

Q7 → B. Row 2 = 2 × row 1 → linearly dependent (low rank) → not invertible. A matrix also cannot be inverted if it is not square.
Why the others are wrong — **A**: Diagonal with non-zero entries → invertible. **C**: Identity — its own inverse. **D**: A permutation (orthogonal) matrix: inverse = transpose.
Look up: W2 p42 · Reference PDF p. 14

Q8 → B. p-value = probability of the observed outcome or something more extreme, computed assuming the null is true. Lower p → more confidence in the alternative.
Why the others are wrong — **A**: A p-value is not the probability that the null is true. **C**: It says nothing about effect size. **D**: It is not a probability of a hypothesis.
Look up: W2 p31 · Reference PDF p. 14

Weak spots to review (tick the questions you missed)

Missed?	Q	Topic	Re-read
☐	Q1	Bayes theorem with total probability	W2 p22, p25 · PDF p. 16
☐	Q2	Independent vs mutually exclusive	W2 p23–p24 · PDF p. 14
☐	Q3	Binomial distribution	W2 p27 · PDF p. 15
☐	Q4	Skewness	W2 p7 · PDF p. 13
☐	Q5	Population vs sample variance	W2 p8 · PDF p. 15
☐	Q6	Eigenvectors and matrix product	W2 p37, p46 · PDF p. 15
☐	Q7	Invertibility	W2 p42 · PDF p. 14
☐	Q8	p-value	W2 p31 · PDF p. 14

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